Abiy W Hailemariam

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Teaching Assistant || AI Researcher || Technical Lead

QUALIFICATION SUMMARY

AI researcher and teaching assistant at Addis Ababa University, focused on model generalization, representation learning, and reliable adaptation of language and multimodal foundation models. My MSc research at the Resonance AI4D Lab investigated evaluation leakage in Amharic legal outcome prediction and precedent retrieval. I bring experience in reproducible machine learning research, university teaching, and engineering leadership.

EDUCATION

BSc, Information Technology March 2022
AAiT, Addis Ababa University Addis Ababa, ETH

MSc, Information and Data Science October 2026
CNCS, Addis Ababa University Addis Ababa, ETH
Advisor: Dr. Martha Yifiru Tachbelie

Thesis: Leakage-Controlled Outcome Prediction and Precedent Retrieval for Amharic Cassation Decisions of the Ethiopian Federal Supreme Court

RELEVANT PROFESSIONAL EXPERIENCE

Teaching Assistant September 2025 - Present
School of Information Science, Addis Ababa University Addis Ababa, ETH

Teach undergraduate AI and systems courses through lectures, practical labs, and mentoring.

- Teach Introduction to AI (2026); previously taught networking and systems administration.
- Lead Linux/RHEL and networking labs covering routing, security, storage, and troubleshooting.
- Design course assessments, grade coursework, and mentor undergraduate students.

Graduate Researcher September 2025 - October 2026
Resonance AI4D Lab, CTBE, AAU Addis Ababa, ETH

Resonance Master's Research Fellow investigating evaluation leakage, Amharic legal outcome prediction, and precedent retrieval Under Dr. Fitsum Assamnew Andargie.

- Curated and audited 7,460 Amharic Cassation decisions; found residual bench reasoning in 73.0% of statistically trimmed cases.
- Evaluated AfroXLMR-large and EthioLLM-large with continued pretraining, tokenizer adaptation, and long-document representations; analyzed representation changes using CKA.
- Compared Transformer encoders and a FastText-BiGRU-Att baseline across leakage-control settings; macro-F1 reached 0.784 with statistical trimming versus 0.499 in a human-trimmed configuration under differing evaluation conditions.
- Developed lexical and dense precedent retrieval using weighted RRF and reranking, achieving 85.8% pool-based Recall@100 against silver relevance judgments.
- Ran multi-seed evaluations with chronological splits, label checks, and human-reviewed relevance judgments.

Technical Lead February 2025 - Present
Vision 5 Tech / Vision Vertex Solutions Addis Ababa, ETH

Lead cross-functional technical delivery across software engineering, cloud/DevOps, Microsoft Power Platform, and AI/data teams.

- Coordinate software delivery, cloud/DevOps, Power Platform, and AI/data workstreams across engineering teams.
- Helped establish engineering teams and development practices; mentor engineers, university students, and recent graduates.
- Deliver AI and software training for school and university students; support professional development within engineering teams.
- Develop an R&D initiative focused on low-resource NLP and applied AI in Ethiopia.

Jr Software Engineer - Sr Software Engineer June 2021 - September 2024
Yaltopia Tech Addis Ababa, ETH

Developed full-stack web and mobile products, supporting system integration, cloud deployments, and team delivery.

- Built and helped deliver production software using React, Node.js, Next.js, Flutter, and cloud technologies.

SELECTED RESEARCH PRESENTATIONS

Deep Learning Indaba 2026 August 2026
Nigeria | Poster Presentation

- Presented encoder-based Amharic Cassation outcome-prediction research and evaluation findings; received a Best Poster Award (one of multiple recipients).